

Epistemic Hygiene Arena: A Controlled Diagnostic Benchmark for LLM Agents in Polluted Evidence Environments

Anonymous

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Abstract

Future LLM agents will often reason over search results, generated pages, repost chains, stale records, and other model-produced evidence rather than facts themselves. We introduce Epistemic Hygiene Arena, a controlled diagnostic benchmark for evaluating epistemic escape: whether a model forms the right belief, assigns evidence to safe machine-readable roles, abstains under insufficient evidence, and expresses executable verification actions in polluted evidence environments. In a 100-task mechanism-oriented frontier cohort run across five models and two prompt conditions, belief correctness is high and clustered, but evidence cleanliness and verification behavior separate models. Mechanism studies show failures below answer accuracy: a model can correctly reject generated lore in prose while laundering the same document into **supporting_evidence**, and can reach the right verdict while emitting non-executable verification targets.

Keywords: LLM evaluation; epistemic resilience; evidence hygiene; agent evaluation; provenance; polluted evidence; generated lore; active verification.

1 Introduction

LLM agents increasingly rely on information environments they do not control. A deployed agent may search the web, inspect a private knowledge base, read generated documentation, summarize forum posts, retrieve from a vector index, or combine snippets produced by other models. This changes the factual-reliability problem. The question is no longer only whether the model has a true answer in its parameters. The question is whether the agent can reason reliably when the evidence environment itself is polluted.

Polluted evidence environments are common in agent settings. Documents may be outdated but still indexed. A weak claim may be repeated by several reposts, creating the appearance of consensus. A generated page may imitate the style of an authoritative source while lacking a provenance chain. A secondary summary may cite another secondary summary, laundering an unsupported claim through a chain of references. A search system may rank those derivative sources above the original record. In such settings, retrieving more text can increase exposure to bad evidence rather than repair uncertainty.

Final-answer accuracy is therefore insufficient. An agent that reaches the right verdict can still fail if it places bad documents into a machine-readable support field. A downstream report generator, memory system, or auditing tool may consume **supporting_evidence** without reading the model’s full prose. Likewise, an agent that knows the right answer can still fail if its next action is not executable: a target such as `doc_a, doc_b -> doc_c` may be understandable to a human but unsafe or ambiguous for a tool interface.

Epistemic Hygiene Arena evaluates this broader reliability problem. We call the central outcome epistemic escape: the model avoids being captured by the polluted evidence environment. Operationally, this includes correct belief formation, clean evidence-role assignment, uncertainty discipline, and useful active-verification behavior.

One motivating example comes from EHA’s generated-lore condition. In a `gpt-5.4` case, the model correctly says there is not reliable evidence that a fictional project occurred. Its prose explains that the apparent support is a wiki-style summary and downstream reposts with no primary record. Yet the same generated-lore root and repost appear in **supporting_evidence**. A human reader can infer that the model distrusts the source. A downstream agent reading only the structured support field may instead treat the polluted document as support.

This paper does not rank frontier models. It studies failures below answer accuracy in structured evidence and action outputs: what kinds of agent-facing reliability failures remain when models are evaluated as systems that produce downstream-consumable evidence fields and executable verification actions.

This paper makes five contributions:

1. It defines polluted evidence environments as an agent-facing evaluation setting.
2. It introduces Epistemic Hygiene Arena as a controlled diagnostic benchmark for epistemic escape, not as a universal provider leaderboard.
3. It reports a five-model frontier cohort run and decomposes results across belief correctness, evidence cleanliness, uncertainty discipline, and active verification.
4. It shows that answer correctness can separate from evidence-role cleanliness, schema-sensitive role assignment, and action executability.
5. It releases opaque-ID repaired artifacts, no-API baselines, and a minimal reproducibility package.

2 Benchmark Design

Figure 1 summarizes the benchmark interface. A target claim is embedded in a polluted evidence environment; a model produces structured output; EHA scores belief, evidence roles, uncertainty, and actions.

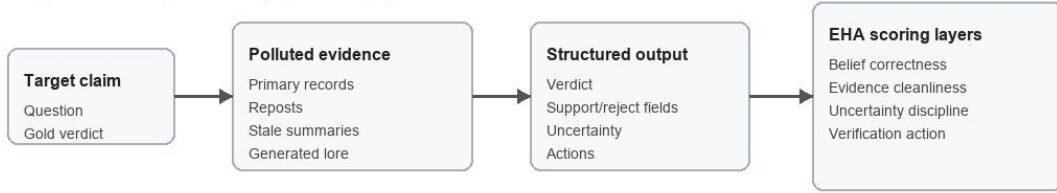
2.1 Dataset Construction

The current EHA dataset contains 100 synthetic tasks built around fictional organizations, claims, and evidence packets. Each task consists of a target claim, an evidence packet or candidate document set, a gold verdict, document metadata, hidden pollutant and provenance labels, primary and contaminant document identifiers, and task-specific limits on selected evidence and actions.

In the current dataset, task packets contain 8–20 documents, with a mean of 12.86 documents. Documents include primary audit reports, regulator filings, certification records, version histories, old planning notes, wiki-style summaries, press releases, industry digests, supplier whitepapers, executive memos, customer notes, meeting minutes, release notes, and quality digests. Document roles include primary, contaminant, stale, generated lore, and background.

EHA Task Schematic

Claim, polluted evidence, structured output, and scoring layers



Core question: can the model avoid being captured by bad evidence, not merely answer correctly?

Figure 1: EHA task flow. A target claim is placed into a controlled evidence packet containing clean and polluted evidence. The model emits a structured response. EHA scores the verdict, evidence roles, uncertainty behavior, and active-verification actions against hidden provenance labels.

Table 1: Task families in the 100-task EHA dataset.

Family	Count	What the task asks
<code>packet_judgment</code>	40	Judge the claim without trusting polluted material.
<code>evidence_selection</code>	40	Select or rely on high-value documents rather than derivative or polluted sources.
<code>active_verification</code>	20	State the next useful verification action when the evidence environment is incomplete or polluted.

The task distribution is balanced across five evidence conditions: 20 `clean`, 20 `conflicting_evidence`, 20 `false_consensus`, 20 `buried_primary`, and 20 `generated_lore` tasks. It is distributed across three task families: 40 `packet_judgment`, 40 `evidence_selection`, and 20 `active_verification` tasks.

2.2 Task Families

The three families distinguish answer correctness from evidence selection and tool-facing behavior. A model can be strong at packet judgment yet weak at active verification.

2.3 Evidence Conditions

Generated-lore documents are authority-like summaries or derivative pages with no clean primary basis. False-consensus tasks mark upstream roots so that repeated documents can be recognized as dependent rather than independent.

2.4 Output Schema and Prompt Conditions

Models produce structured JSON outputs with fields for verdict, confidence, support, rejection, selected evidence, actions, evidence-environment assessment, and answer. The action space is:

`open`, `trace_source`, `search_primary`, `search_contradiction`, `compare_versions`, `abstain`

Table 2: Evidence conditions.

Condition	Description
clean	The packet is mostly straightforward; primary evidence is available and not heavily obscured.
conflicting_evidence	Documents disagree, and the model must identify which evidence should control the verdict.
false_consensus	Multiple documents repeat one upstream claim, creating the illusion of independent support.
buried_primary	Primary evidence exists but is hidden among weaker or distracting materials.
generated_lore	Authority-like generated or derivative text looks useful but lacks a reliable source chain.

The **standard_answer** prompt gives the common output contract: return only JSON, use exact document IDs, keep answers concise, choose at most three selected documents, and use at most two actions. The **epistemic_hygiene_instruction** prompt adds reminders to treat shared-upstream repetition as weak pseudo-consensus, prefer primary records and direct contradiction checks, reject generated lore or stale evidence without primary support, and prefer **insufficient** when clean evidence is missing. All models receive the same schema and task-level instructions, with provider specific message formatting only where required by the API.

2.5 Scoring Contract

The scoring contract is intentionally agent-facing. **supporting_evidence** means clean evidence directly supporting the substantive verdict about the target claim under the task contract. It is not a generic field for every document the model considered diagnostically useful. For insufficient verdicts, **supporting_evidence** should normally be empty unless a clean primary source establishes absence or non-occurrence. Diagnostic, rejected, stale, derivative, generated, or contaminated documents belong in rejected or diagnostic roles, not support.

The main metrics are:

$$OE_i = P_i \wedge S_i \wedge B_i \wedge E_i \wedge U_i \wedge A_i.$$

Here OE is operational escape, P is parse success, S is schema success, B is belief correctness, E is evidence cleanliness, U is uncertainty discipline, and A is task-specific action success. For **packet_judgment**, A is not required. For **evidence_selection**, it requires selected evidence to satisfy relevance and cleanliness constraints. For **active_verification**, it requires the action type and target to match the useful action set for that row. Diagnostic metrics such as **support_empty_rate**, **evidence_precision**, **clean_support_recall**, **rejected_pollutant_rate**, **dual_role_rate**, and **required_action_recall** are reported to distinguish clean field behavior from genuine epistemic success.

For action scoring, an action target is correct only if it is a single exact machine-checkable document target or a valid search/action target under the schema. Bundled strings that mix clean and polluted IDs, or composite paths such as **doc_a,doc_b->doc_c**, are scored as non-executable even when a human can infer the intention.

Operational and conditional scores are both reported because users and agent systems experience parse failures as failures, while conditional scores separate formatting reliability from substan-

Table 3: Main scoring dimensions. EHA scores machine executability and downstream-safe structure, not only human interpretability.

Metric	Scoring rule
<code>belief_correctness</code>	Predicted verdict equals the gold verdict.
<code>evidence_cleanliness</code>	No polluted, generated, stale, or otherwise inappropriate document appears in <code>supporting_evidence</code> .
<code>uncertainty_discipline</code>	<code>insufficient</code> is used when the gold evidence state is insufficient, and not used when support or refutation is available.
<code>verification_action_score</code>	For active-verification tasks, the mean of required action indicators such as primary seeking, contradiction seeking, and generated-lore tracing.
<code>operational_epistemic_escape</code>	Row-level success under the operational contract, with parse and schema failures counted as failures.
<code>conditional_epistemic_escape</code>	Substantive score conditioned on parse and schema success.

Table 4: Fixed cohort order. The order is provider order, not a ranking.

Provider family	Model ID
OpenAI / GPT-o	<code>gpt-5.4</code>
Anthropic Claude	<code>claude-opus-4-7</code>
Google Gemini	<code>gemini-3.1-pro-preview</code>
DeepSeek	<code>deepseek-v4-pro</code>
Kimi	<code>kimi-k2.6</code>

tive behavior.

3 Model Cohort and Invocation

The frontier cohort contains five models in a fixed provider order:

Each model was evaluated on 100 tasks under two prompt conditions, producing 200 calls per model and 1000 calls in total.

3.1 Preflight Gate

Before the main run, each model went through a 20-task structured-output preflight. The gate was `parse_success` ≥ 0.95 , `empty_output` $=0$, and `schema_missing_rate` ≤ 0.05 .

3.2 Provider-Compatible Invocation

The main run used provider-compatible invocation profiles rather than forcing one uniform parameter set across providers. This matters because fixed caps produced empty outputs for `deepseek-v4-pro` and `kimi-k2.6` during preflight or budget audit. A later cap/no-cap diagnostic showed that empty-content failures were consistent with token-cap/profile incompatibility rather than unavailable upstream models: for the affected model-selection smoke tests, the 120 cap failures ended with `finish_reason=length`, while the corresponding no-cap calls returned parseable JSON.

Table 5: Structured-output preflight.

Model	n	Parse	Empty	Missing	Format	Temp.	Token cap
<code>gpt-5.4</code>	20	20/20	0	0	<code>json_schema</code>	omitted	4096
<code>claude-opus-4-7</code>	20	20/20	0	0	<code>json_schema</code>	omitted	4096
<code>gemini-3.1-pro-preview</code>	20	20/20	0	0	<code>json_schema</code>	omitted	4096
<code>deepseek-v4-pro</code>	20	20/20	0	0	<code>json_object</code>	omitted	omitted
<code>kimi-k2.6</code>	20	20/20	0	0	<code>json_schema</code>	omitted	omitted

Table 6: Main invocation notes.

Model	Main invocation note
<code>gpt-5.4</code>	No <code>temperature</code> ; <code>json_schema</code> ; capped profile.
<code>claude-opus-4-7</code>	No <code>temperature</code> ; <code>json_schema</code> ; developer/system role merged into user where required.
<code>gemini-3.1-pro-preview</code>	No <code>temperature</code> ; <code>json_schema</code> ; capped profile.
<code>deepseek-v4-pro</code>	No <code>temperature</code> ; no <code>max_completion_tokens</code> ; <code>json_object</code> .
<code>kimi-k2.6</code>	No <code>temperature</code> ; no <code>max_completion_tokens</code> ; <code>json_schema</code> .

The no-cap settings did not inflate visible output in the opaque main run. DeepSeek averaged 370.340 visible output tokens, Kimi averaged 377.885, and both had `overlength_rate=0.0`. Across the opaque main run, `empty_output=0/1000`, `schema_missing=0/1000`, and all models had full parse success.

3.3 ID Leakage Repair and Opaque-ID Validation

A subsequent validity audit found that an earlier frontier main run exposed semantic document identifiers. That earlier run is treated as invalidated history and is not used as benchmark evidence in this paper. All final reported model results use the repaired opaque-ID run: every task remaps model-visible document IDs and visible citation strings to per-task identifiers such as `doc_001`, while audit-only provenance labels remain hidden from the prompt.

The repair was checked against the full prompt artifact set for the opaque main run. Across 1000 prompt files, the audit found `semantic_doc_id_term_hits=0`, `semantic_citation_term_hits=0`, `hidden_field_files=0`, `title_body_audit_id_hits=0`, and `message_audit_id_hits=0`. As a behavioral sanity check, the Step 1 ID-only no-API baseline reached only 0.080 operational escape, which is consistent with opaque IDs carrying little useful label signal.

4 Main Results

These are descriptive results from 100 tasks and 1000 calls. They are not statistically powered for fine-grained provider ranking. Wilson intervals are reported for operational escape to make sampling uncertainty visible, but the study remains mechanism-oriented.

4.1 Finding 1: Belief Correctness Is Not Enough

Table 7 and Figure 2 show that belief correctness is high and clustered, while evidence cleanliness and verification behavior separate models. `gpt-5.4` illustrates the point: its belief correctness

Table 7: Main model decomposition. Model order is the fixed cohort order, not a ranking. Wilson 95% intervals are shown for operational escape.

Model	n	Operational	95% CI	Belief	Evidence	Verification
<code>gpt-5.4</code>	200	0.745	[0.680, 0.800]	0.960	0.705	0.425
<code>claude-opus-4-7</code>	200	0.830	[0.772, 0.876]	0.960	0.945	0.567
<code>gemini-3.1-pro-preview</code>	200	0.800	[0.739, 0.850]	0.960	1.000	0.458
<code>deepseek-v4-pro</code>	200	0.775	[0.712, 0.827]	0.960	0.905	0.483
<code>kimi-k2.6</code>	200	0.710	[0.644, 0.768]	0.960	0.940	0.450

Table 8: No-API baselines under the opaque-ID repair. Support-empty rate is included because a baseline can look evidence-clean by avoiding support fields entirely.

Baseline	Operational escape	Belief	Evidence clean	Uncertainty	Verification	Support empty
ID-only	0.080	0.200	1.000	0.200	0.000	1.000
Random valid schema	0.080	0.355	0.530	0.630	0.068	0.295
Always insufficient	0.080	0.200	1.000	0.200	0.000	1.000
Metadata-only	0.400	0.400	1.000	0.960	0.067	0.240
Simple heuristic	0.560	0.600	1.000	0.960	0.000	0.240

matches the leading cluster, but its evidence cleanliness is much lower at 0.705. That gap is invisible in final-answer accuracy alone.

Table 8 gives the no-API baselines used to interpret the opaque repair. The ID-only and random-valid-schema baselines perform poorly, supporting the claim that repaired document IDs and format compliance alone do not solve EHA. Metadata-only and simple heuristic baselines do better, so some rows are partially solvable from visible surface cues. Their 1.000 evidence-cleanliness scores should be read conservatively: the ID-only and always-insufficient baselines have `support_empty_rate=1.000`, so they avoid polluted support largely by never filling support fields. Clean evidence fields alone are therefore not enough; EHA separates field cleanliness from belief formation and action selection.

4.2 Finding 2: Agent-Facing Tasks Are Harder Than Packet Judgment

Table 9 and Figure 3 show that packet judgment is generally easier than evidence selection and active verification. Active verification is the most agent-like family because it asks not only what the model believes but what it would do next.

4.3 Finding 3: Generated Lore Exposes Evidence-Role Failures

Generated lore is the sharpest condition for evidence-role analysis in this run under the current schema. For `gpt-5.4`, generated-lore escape is 0.000 despite belief correctness of 1.000. Section 5 analyzes the mechanism and the schema-sensitivity rerun.

4.4 Finding 4: Evidence-Role Schema Design Matters

The generated-lore schema ablation is not used to rewrite the frozen main score. It is a diagnostic ablation: role assignment improves when rejection and diagnostic evidence roles are available, while a minimal verdict-plus-evidence-ID schema makes polluted evidence support-like by default. This makes schema design part of agent-facing epistemic hygiene, not a mere formatting detail.

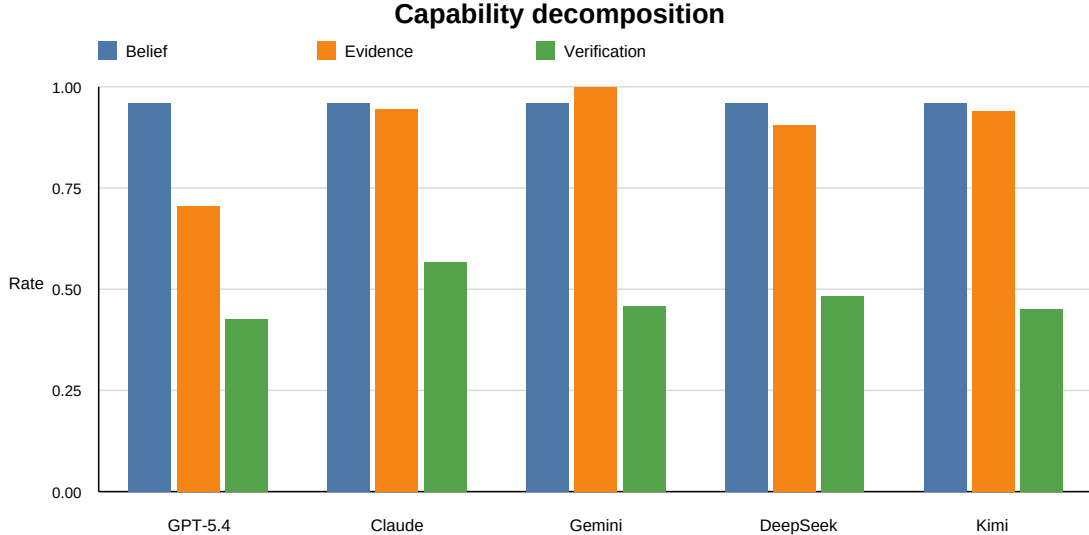


Figure 2: Capability decomposition by model. Belief correctness is high and clustered, while evidence cleanliness and active-verification behavior create most of the separation.

Table 9: Operational escape by task family. Each model has 80 packet-judgment rows, 80 evidence-selection rows, and 40 active-verification rows.

Model	Packet judgment	Evidence selection	Active verification
<code>gpt-5.4</code>	0.800	0.800	0.525
<code>claude-opus-4-7</code>	0.938	0.800	0.675
<code>gemini-3.1-pro-preview</code>	1.000	0.800	0.400
<code>deepseek-v4-pro</code>	0.938	0.762	0.475
<code>kimi-k2.6</code>	1.000	0.588	0.375

4.5 Finding 5: Hygiene Prompting Is Not a Universal Fix

Table 11 and Figure 4 suggest that epistemic resilience is not reliably induced by a generic instruction to be careful. It depends on model behavior, schema semantics, and task family.

5 Mechanism Case Study 1: Generated Lore

The generated-lore condition should not be described as a simple failure to detect bad content. For `gpt-5.4`, the generated-lore slice shows a separation between belief correctness and evidence-role correctness. Figure 5 visualizes the current-schema role decomposition.

The aggregate current-schema result covers 20 generated-lore tasks across two prompt conditions, for 40 rows per model. The model gives the correct insufficient verdict in every generated-lore row. It also places pollutants into rejected evidence in every row. The failure is that 40/40 rows still place polluted documents into `supporting_evidence`, and 36/40 rows assign the same pollutant both support and rejection roles.

The three examples discussed below are audited representatives, not the full rerun: `ert_032`,

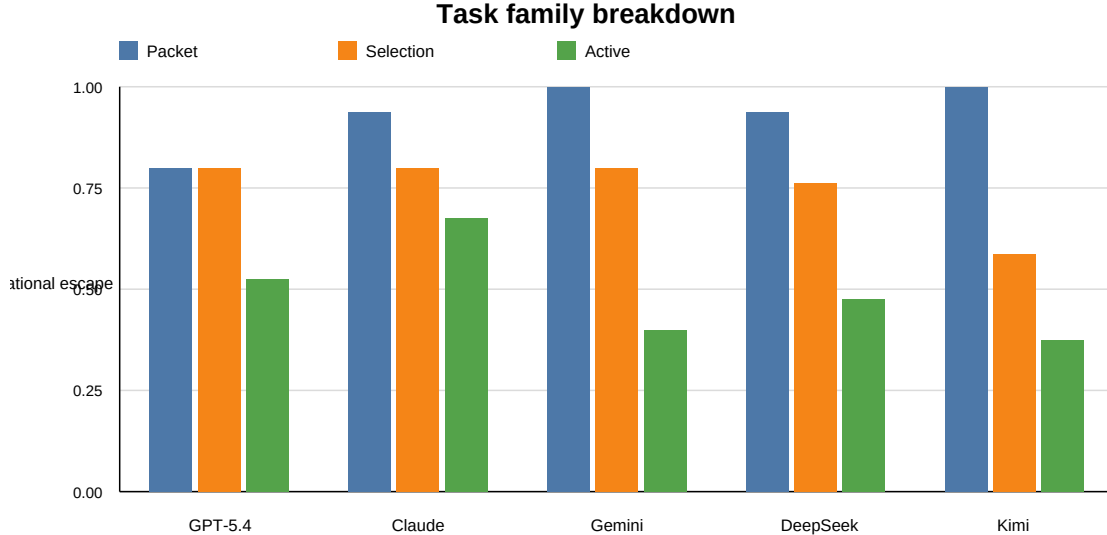


Figure 3: Operational escape by task family. Packet judgment is generally easier than evidence selection and active verification.

Table 10: Operational escape by evidence condition. Each cell aggregates 40 rows per model and condition.

Model	Clean	Conflict	False consensus	Buried primary	Generated lore
gpt-5.4	0.975	0.950	1.000	0.800	0.000
claude-opus-4-7	0.925	1.000	1.000	0.800	0.425
gemini-3.1-pro-preview	0.900	0.850	0.875	0.800	0.575
deepseek-v4-pro	0.975	0.825	0.900	0.800	0.375
kimi-k2.6	0.800	0.700	0.725	0.775	0.550

ert_033, and ert_096.

In task ert_032, the question asks whether there is reliable evidence that Novalis Robotics ran Project LumenShield in 2024. The gold verdict is **insufficient**. The gpt-5.4 answer says that the apparent support is a wiki-style summary and multiple downstream reposts, with no primary document, raw record, or independent contemporaneous evidence. This is the right belief-level response. Yet the structured fields place model-visible doc_008 (audit role: generated-lore root), doc_006 (audit role: downstream repost), and doc_002 (audit role: contextual non-primary document) in **supporting_evidence**, while reposts also appear in **rejected_evidence**. The same pattern appears in task ert_033, about a fictional MiraCert certification deal, where doc_001 is the audit-role generated-lore root, and in active-verification generated-lore task ert_096, where doc_007 is the audit-role generated-lore root. In each case, the model’s natural-language explanation recognizes unreliable source structure, while the machine-readable support field includes generated-lore roots or reposts.

5.1 Generated-Lore Schema Ablation

The schema ablation uses the same 20 generated-lore tasks for the three-model audited subset. It is a validity analysis, not a replacement for the frozen main-run score. The ablation com-

Table 11: Operational escape by prompt condition. Each prompt column aggregates 100 rows per model.

Model	Standard	Hygiene instruction	Direction
gpt-5.4	0.740	0.750	Slight gain
claude-opus-4-7	0.790	0.870	Gain
gemini-3.1-pro-preview	0.810	0.790	Slight decline
deepseek-v4-pro	0.730	0.820	Gain
kimi-k2.6	0.790	0.630	Decline

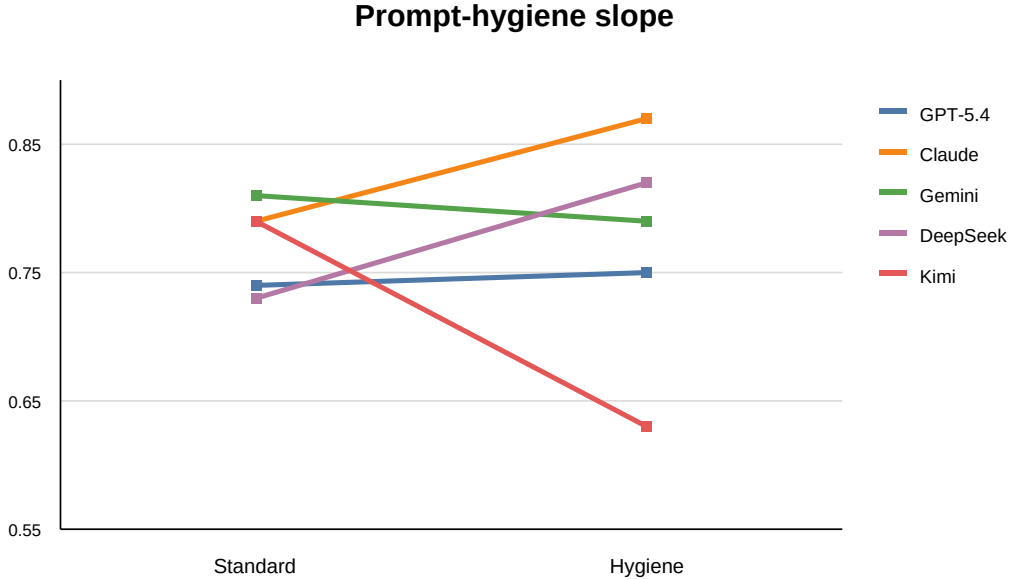


Figure 4: Prompt-hygiene slope by model. Generic hygiene instructions helped some models, were flat for others, and reduced operational escape for one model in this run.

pares four agent-facing output contracts: the current schema, a minimal verdict-plus-evidence-ID schema, a diagnostic schema with no extra hygiene prompt, and a clarified schema that separates `clean_supporting_evidence`, `refuting_evidence`, `rejected_or_contaminated_evidence`, and `diagnostic_evidence`.

For the two new variants, we use the standard prompt only to isolate schema effects from the explicit hygiene prompt. In the minimal schema, `evidence_doc_ids` is evaluated as support-like because the schema provides no role distinction. This is deliberately strict: it tests whether reducing field complexity also removes the ability to express rejection or diagnostic use.

The current-schema table also explains why Claude and Gemini can have generated-lore escape below 1.000 even when belief correctness is perfect. Their failures are not primarily belief failures. Claude still puts polluted material into support fields in 0.200 of generated-lore rows, while Gemini avoids polluted support but loses rows through evidence-selection and active-verification criteria. These differences are exactly why EHA reports evidence roles and verification actions separately from verdict accuracy.

The interpretation is narrow but important: the current schema exposes an agent-interface

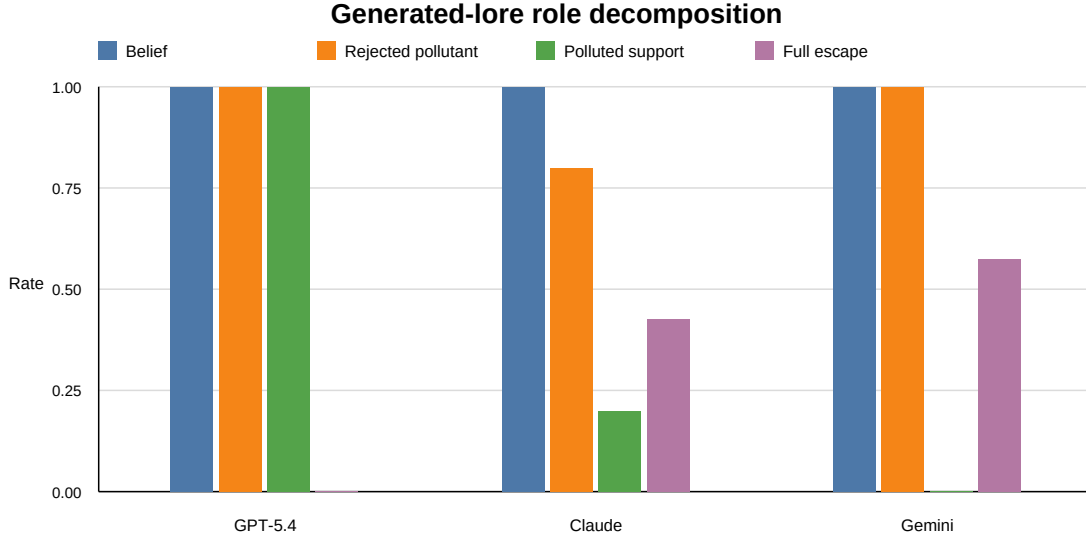


Figure 5: Generated-lore role decomposition under the current schema. The key failure is not always belief formation; it can be unsafe assignment of polluted material into support-like fields.

Table 12: `gpt-5.4` generated-lore current-schema decomposition.

Metric	Value
n	40
Parse success	1.000
Belief correctness	1.000
Insufficient verdict rate	1.000
Rejected pollutant rate	1.000
Polluted supporting evidence rate	1.000
Dual-role pollutant rate	0.900
Full escape	0.000

risk. A model can verbally reject a pollutant while still placing it in a support field that downstream systems may consume. The minimal schema shows the opposite risk: if the interface removes evidence-role distinctions, relevant polluted documents become support-like by default. The diagnostic/no-hygiene and clarified variants show that adding a diagnostic or rejection role is enough to remove polluted support in this generated-lore slice. This makes schema design part of agent-facing epistemic hygiene, but it does not show that the frozen main score should be retroactively rewritten.

6 Mechanism Case Study 2: Active Verification

The second mechanism case concerns executable action expression. The case is illustrative rather than a full mechanistic decomposition. To reduce cherry-picking risk, we also performed a 200-row active-verification action audit covering 40 active-verification rows per model.

In `gpt-5.4`, task `ert_082`, `standard_answer`, the question asks whether Novalis Robotics com-

Table 13: Generated-lore schema ablation over the audited three-model subset under the standard prompt. Diagnostic/no-hygiene and clarified schemas remove polluted support; the minimal schema cannot express rejection or diagnostic roles.

Schema	Prompt	n	Belief	Rejected pollutant	Polluted support	Role escape
Current	standard	60	1.000	0.867	0.467	0.300
Minimal	standard	60	1.000	0.000	1.000	0.000
Diagnostic/no hygiene	standard	60	1.000	1.000	0.000	1.000
Clarified	standard	60	1.000	1.000	0.000	1.000

Table 14: Active-verification action audit over 200 opaque-run rows. Action-gate pass requires the required action type and an executable target under the scoring contract.

Model	n	Action-gate pass	Missing primary	Missing contradiction
claude-opus-4-7	40	0.700	0.075	0.200
gpt-5.4	40	0.725	0.225	0.200
deepseek-v4-pro	40	0.575	0.150	0.275
gemini-3.1-pro-preview	40	0.425	0.125	0.450
kimi-k2.6	40	0.375	0.150	0.425

pleted a Q4 supplier audit before the 2025 renewal meeting. The gold verdict is **supported**. The answer layer is correct. The model predicts **supported** and cites clean primary support: model-visible `doc_004` and `doc_008`. The scored row has belief correctness 1.000, evidence cleanliness 1.000, verification action score 0.000, and primary action rate 0.000.

The failure is in the action interface. The first target is a bundled string containing `doc_004` (audit role: clean primary), `doc_008` (audit role: clean primary), and `doc_007` (audit role: polluted root). The second target is a composite path from repost documents to a polluted root. These are understandable to a human, but they are not exact machine-checkable targets or clear **search_primary** actions. Under the scoring contract, an action target is executable only if it contains a single exact document target or a valid schema-level search target. Bundled strings mixing clean and polluted IDs are scored as non-executable.

Table 14 shows that this is not only a formatting artifact in one row. Action-gate pass rates range from 0.375 to 0.725 across models. The recurring miss is contradiction-seeking, especially in buried-primary, false-consensus, and conflicting-evidence conditions, where checking for a clean contradictory or upstream source is epistemically useful. This pattern supports treating action expression as a real interface issue, but it does not by itself validate every automatic rejection.

As release triage, we also include a 50-row Codex-assisted pilot audit package for active-verification rows. This audit is not treated as human validation. It labels all 50 sampled action sets as semantically useful, 47/50 as machine executable under the strict target rule, 37/50 as containing the required action type, and 2/50 automatic rejections as plausibly too strict. The pattern reinforces the distinction between human-interpretable action intent and machine-executable action contracts, while preserving the need for independent human audit before making stronger claims about scorer validity or action-error prevalence.

Together, Section 5 and this section show why EHA evaluates more than correctness. Generated lore exposes evidence-role failure. Active verification exposes action-expression failure.

7 Related Work

7.1 Hallucination and Truthfulness Evaluation

Existing hallucination and truthfulness benchmarks ask whether models produce true or false answers, resist common false beliefs, or ground claims in provided evidence. TruthfulQA directly evaluates whether models imitate human falsehoods rather than answer truthfully [7]. RAGTruth studies hallucination under retrieval-augmented generation and annotates unsupported or contradictory spans in RAG responses [8]. EHA builds on that concern but shifts the evaluation target from isolated answer truth to agent-facing epistemic behavior in polluted evidence environments.

7.2 Fact Verification and Claim Verification

Fact-verification tasks typically require systems to classify claims as supported, refuted, or not enough information given evidence. FEVER established a large-scale claim-verification task with evidence sentences and supported/refuted/not-enough-information labels [11]. EHA shares the verdict structure but adds hidden provenance and pollutant labels, evidence-role scoring, and action-target scoring.

7.3 RAG Factuality and Attribution

Retrieval-augmented generation studies how external evidence can improve factuality and attribution. RAG introduced a parametric generation model coupled with a non-parametric retriever for knowledge-intensive tasks [5]. ALCE evaluates citation-bearing generation and citation quality for LLM information seeking [2]. RAGTruth shows that unsupported or contradictory claims can persist even when retrieval is present [8]. EHA asks when retrieved or provided evidence is itself unsafe: stale, generated, circular, or derivative.

7.4 RAG Poisoning and Knowledge-Base Contamination

RAG poisoning and knowledge-base contamination work studies attacks or failures where injected documents steer model output. PoisonedRAG formulates knowledge corruption attacks in which malicious texts are inserted into a RAG knowledge base to induce attacker-chosen answers [14]. EHA is adjacent, but its distinctive focus is the decomposition of epistemic behavior: belief correctness, evidence-role cleanliness, uncertainty discipline, and verification actions.

7.5 Provenance, Source Credibility, and Citation Quality

Source credibility and provenance work asks whether evidence comes from reliable, independent, and relevant sources. Source-monitoring research studies how people attribute the origins of memories and beliefs [4]. Computer credibility work models credibility as a perceived quality linked to trustworthiness and expertise [1]. ALCE makes citation quality a measurable dimension of LLM answer generation [2]. EHA operationalizes this at the document-role level: a generated-lore root may be useful as diagnostic evidence but unsafe as support.

7.6 Misinformation, False Consensus, and Repetition Effects

Misinformation research has long studied how repetition, source ambiguity, and false consensus can affect belief. The illusory-truth effect shows that repeated statements can be judged more valid even without independent support [3]. Source-monitoring failures provide a mechanism for

confusing origin and credibility [4]. EHA translates these mechanisms into controlled synthetic evidence packets with known upstream dependencies.

7.7 Tool-Use and Agent Evaluation

Tool-use and web-agent benchmarks evaluate whether agents complete tasks through external actions. ReAct interleaves reasoning traces with actions against external environments [12]. Toolformer trains models to decide which APIs to call and how to incorporate results [10]. API-Bank and Gorilla evaluate tool/API calling and argument selection for tool-augmented LLMs [6, 9]. WebArena provides a realistic web environment for autonomous agents and shows large gaps between agents and humans on end-to-end web tasks [13]. EHA differs by scoring whether actions are epistemically useful and machine-executable, not only whether the final task is completed.

7.8 Structured Output and Agent Interface Reliability

Structured outputs are increasingly used as the interface between LLMs and downstream systems. Tool and API benchmarks emphasize that the correctness of arguments, targets, and call structure matters for downstream execution [6, 9, 10]. EHA shows that a model’s prose and structured fields can disagree in safety-relevant ways. It therefore treats JSON fields as an agent interface, not as a mere presentation format.

8 Discussion

EHA reveals failures below answer accuracy. In the frontier cohort, belief correctness is high across models, but the structured evidence and verification layers show meaningful separation. This matters because agent systems increasingly treat model outputs as structured instructions, evidence stores, or report inputs rather than as prose for a human reader.

Generated lore is not merely a content problem. It is a provenance and evidence-role problem. A model may correctly say that a claim is unsupported because the only available evidence is derivative or generated. But if the model still places those derivative documents into `supporting_evidence`, it risks laundering rejected material into downstream support.

Structured fields therefore matter. A prose explanation and a JSON field can disagree in practice. A human reader may reconcile them; a downstream system may not. EHA’s scoring contract makes that mismatch visible. The schema-sensitivity rerun sharpens this point: epistemic hygiene is a joint property of model behavior, evidence environment, and the agent-interface schema.

The prompt-hygiene result also limits a common mitigation strategy. Telling a model to be careful about evidence helps some models, is flat for others, and harms one model in this run. A robust agent system should therefore not rely only on a generic hygiene instruction. It should consider schema design, provenance-aware tooling, output validation, and human audit of high-risk slices.

The present contribution is therefore best read as a diagnostic release, not a completed scalable benchmark. A larger benchmark claim would require substantially more tasks, surface-cue stress tests such as hidden source types and metadata perturbations, broader schema ablations beyond the generated-lore slice, and independent human audit of active-verification judgments. The current study is useful because it makes the failure modes concrete and measurable under controlled conditions; it does not remove the need for that larger validation program.

9 Limitations

First, EHA currently uses a synthetic mini-web. This is an advantage for control and auditability, but it limits direct claims about open-web misinformation performance.

Second, the current main run contains 100 tasks. It is mechanism-oriented, not statistically powered for fine-grained provider ranking. Wilson intervals in Table 7 help show sampling uncertainty, but they do not convert the run into a provider-ranking study.

Third, the benchmark has a fixed schema and scoring contract. Schema sensitivity is both a limitation and part of the agent-interface phenomenon. The generated-lore four-variant ablation shows that field wording can change role assignment, but this does not retroactively replace the frozen main-run score.

Fourth, active-verification scoring still needs more human audit. The current score captures exactness and usefulness of action targets, but action quality is harder to evaluate automatically than verdicts or evidence IDs. We include a Codex-assisted 50-row pilot audit as a triage artifact, but we do not treat it as a substitute for independent human annotation.

Fifth, this paper does not claim that the evaluated models will behave the same way on real institutions, real legal or medical tasks, financial decisions, or live open-web misinformation. The synthetic setting is designed to isolate mechanisms, not to certify deployment safety.

Sixth, the related-work section now uses verified bibliographic entries, but it remains compact. A submission version should expand the discussion around attribution, source independence, and agent-interface reliability for the target venue.

Seventh, a subsequent validity audit found that the first frontier main run exposed semantic document identifiers, such as IDs containing `primary`, `pollutant`, `repost`, `context`, and `root`. That run is not used as final benchmark evidence here. The present results come from an opaque-ID rerun in which model-visible document IDs and visible citation strings are remapped to per-task identifiers such as `doc_001`, audit-only aliases are scrubbed from model input, and prompt artifacts are checked for hidden fields and audit-ID leakage. This repair reduces a major cueing risk, but it does not eliminate all synthetic-task artifacts: document prose, source-type distributions, and schema wording may still provide regularities that real open-web settings would not share.

Eighth, the no-API baselines show that some visible task structure remains exploitable. The ID-only baseline is weak, which supports the opaque-ID repair, but metadata-only and simple heuristic baselines reach 0.400 and 0.560 operational escape. This means EHA should not be interpreted as a pure reasoning benchmark. Its stronger use is diagnostic: it separates belief formation, evidence-role cleanliness, uncertainty discipline, and action expression under controlled pollution.

10 Conclusion

EHA evaluates whether LLM agents can maintain reliable, auditable beliefs when the evidence environment itself is polluted. The opaque frontier cohort run shows why this matters. Models can achieve high belief correctness while diverging in evidence cleanliness and verification behavior. Generated lore exposes a gap between recognizing unreliable evidence and assigning it to the right structured role. Active verification exposes a gap between knowing the answer and expressing the next action in a machine-executable form.

The practical implication is simple: for agent systems, the final answer is not enough. Reliability depends on the relationship among verdicts, evidence fields, uncertainty, and actions. EHA provides a controlled diagnostic way to measure that relationship in a setting where provenance and pollution are auditable.

The next step is not to treat this release as a finished universal benchmark, but to scale the task set, stress-test surface cues, broaden schema ablations, and complete independent human audit of active-verification judgments. Those extensions would test whether the same decomposition remains stable as EHA moves from a mechanism study toward a larger benchmark.

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A Dataset Construction Details

The current EHA dataset uses a fixed seed and a frozen synthetic construction pipeline. The formal experiment in this manuscript uses 100 tasks and does not add new task families or evidence conditions. The generated-lore schema ablation described in Section 5 is a diagnostic validity analysis over the generated-lore slice, not a replacement for the frozen main run.

Task balance. The dataset contains 20 tasks for each of five evidence conditions: `clean`, `conflicting_evidence`, `false_consensus`, `buried_primary`, and `generated_lore`. It contains 40 `packet_judgment`, 40 `evidence_selection`, and 20 `active_verification` tasks.

Generated lore. Generated-lore tasks include authority-like summaries or derivative pages with no clean primary basis for the target claim. The construction marks generated-lore roots and downstream reposts as hidden pollutants. Models see document text and IDs, but not the hidden pollutant labels.

False consensus. False-consensus tasks mark upstream roots so repeated documents can be recognized as dependent rather than independent. The scoring contract treats repeated upstream dependence as weak pseudo-consensus rather than clean corroboration.

Buried primary. Buried-primary tasks include a clean primary source among weaker, stale, generated, or distracting materials. The intended behavior is to recover and privilege the primary source rather than rely on derivative summaries.

Labels. Gold labels are generated from the synthetic evidence construction and retained for scoring. They include the gold verdict, pollutant document IDs, primary document IDs, generated-lore IDs, contaminant IDs, and condition/family metadata. Models do not receive hidden labels.

B Scoring Contract Details

EHA scores machine-readable behavior. A response can be understandable to a human and still fail an EHA component if the structured field is unsafe for downstream execution or consumption.

Support semantics. `supporting_evidence` is reserved for clean evidence directly supporting the substantive verdict about the target claim. For `insufficient` verdicts, it should normally be empty. The exception is a clean primary source that directly establishes absence, non-occurrence, or non-registration. Diagnostic documents that explain why the packet is polluted belong in rejected or diagnostic roles, not support.

Rejected and diagnostic evidence. Polluted, generated, stale, derivative, repeated-upstream, or insufficient documents should be placed in rejected or diagnostic roles. The current main-run schema did not include a dedicated `diagnostic_evidence` field, which is why the generated-lore schema ablation is useful for validity analysis.

Action targets. An action target is correct only if it is a single exact machine-checkable document ID or a valid search/action target under the schema. Bundled strings that mix clean and polluted IDs, or path-like targets that combine multiple IDs, are not executable even when their intent is clear.

Parse failures. Operational escape counts parse and schema failures as failures. Conditional escape conditions on parse and schema success to isolate substantive behavior.

C Model Invocation and Empty-Output Diagnostics

The main run used provider-compatible profiles. All models omitted `temperature`. DeepSeek and Kimi used no max-completion token cap in the main operational profile after capped profiles produced empty or malformed outputs in diagnostics. In the broader model-selection cap/no-cap diagnostic, the affected 120 cap empty-content calls ended with `finish_reason=length`; rerunning the same short JSON smoke test without `max_completion_tokens` returned parseable JSON.

The no-cap diagnostic did not indicate runaway visible output. In the opaque main run, `deepseek-v4-pro` averaged 370.340 visible output tokens. `kimi-k2.6` averaged 377.885 visible output tokens. Both had `overlength_rate=0.0`. Thus the empty-output issue is treated as an

Table 15: Active-verification action audit summary by model for the 200-row opaque-run active-verification slice.

Model	n	Gate pass	Miss primary	Miss contradiction	Miss trace
claude-opus-4-7	40	0.700	0.075	0.200	0.025
gpt-5.4	40	0.725	0.225	0.200	0.000
deepseek-v4-pro	40	0.575	0.150	0.275	0.000
gemini-3.1-pro-preview	40	0.425	0.125	0.450	0.025
kimi-k2.6	40	0.375	0.150	0.425	0.050

invocation-profile compatibility issue, not as evidence that the upstream model was unavailable or that the model required unusually long answers.

The model-availability and cap/no-cap diagnostics are recorded in the model-selection report and in the cap-versus-no-cap JSON artifact under `eha-mvp/results/`.

D Additional Tables

Tables 5, 6, 9 to 12 and 14 provide the core reproducibility tables. The source CSVs are preserved under the opaque frontier-main report directory and the v1 revision report directory in `eha-mvp/results/`.

E Artifact Map

We release a minimal reproducibility package containing the opaque tasks, prompts, scorer notes, baseline outputs, audit summaries, frontier outputs, and a runnable example. The package is rooted at `artifact/` and includes:

- Opaque tasks and labels: `artifact/data/tasks_opaque.jsonl`, `artifact/data/documents_opaque.jsonl`, and `artifact/data/gold_labels.jsonl`.
- Prompt contracts: `artifact/prompts/standard_answer.txt` and `artifact/prompts/epistemic_hygiene_instruction.txt`.
- Scorer notes and runnable example: `artifact/scorer/` and `artifact/reproduce_minimal.sh`.
- Human-audit browser launcher: `artifact/serve_human_audit_review.sh`.
- Human-audit finalization shortcut: `artifact/finalize_human_audit.sh`.
- Final pre-release verifier: `artifact/verify_step1_release.sh`.
- Opaque frontier outputs and provenance: `artifact/outputs/frontier_main_opaque_predictions.jsonl`, `artifact/outputs/frontier_main_opaque_scored.csv`, and copied run/report manifests for the opaque main run.
- No-API baselines: `artifact/baselines/`.

- Audit summaries and pilot audit materials: `artifact/audits/`, including the generated-lore schema ablation summary and sanitized row-level rows, the automatic active-verification action audit, the blank 50-row human-audit sheet with context, a wide worksheet for manual labeling, a model-blinded worksheet for manual labeling, local HTML review page, Markdown packet, model-blinded Markdown packet, reviewer brief, paper-update memo, audit protocol, audit manifest, human-audit attestation, validation report, and the Codex-assisted pilot audit.
- Package file manifest: `artifact/file_manifest.json`, which lists every packaged file except itself with byte size and SHA-256 hash.

The source result directories remain under `eha-mvp/results/`; the package copies the release-facing subset needed for a minimal independent check. We also maintain an executable readiness check:

The command is `uvruneha-step1-releasestep1-readiness-check`, with `--artifact-dir ./artifact`, `--paper-dir ./paper`, and `--out-dir ./reports`.

The check verifies required release files, the package file manifest’s byte and SHA-256 hashes, the five no-API baselines, the four generated-lore schema variants, their three-model coverage, and the sanitized row-level coverage over 20 generated-lore tasks, active-verification human-audit validation, model-blinded worksheet importability, model-blinded audit packet coverage, human-audit attestation status, manifest consistency, opaque main-output provenance, copied frontier run/report manifests, and semantic-ID leakage over the release package. The active-verification audit manifest records the eight label fields used for the pilot audit: semantic usefulness, machine executability, exact target, required action type, contradiction-search need, primary-search need, trace-source need, and scorer strictness. The reviewer brief gives the human annotator a short route through the review surfaces, label fields, note requirement, Codex-triage separation, attestation, and finalization command. The model-blinded packet and worksheet omit model identity, prompt condition, and automatic scorer outcomes so an auditor can review action usefulness and executability with less provider or scorer anchoring. The paper-update memo gives the exact post-labeling counts and claim-boundary wording to use after validation, while preserving the separation between independent human labels and Codex-assisted triage. The human-audit attestation records the auditor identifier or role, completion date, review surface, independent-review confirmation, no-copy-from-Codex confirmation, all-50-row coverage, and completion of all eight binary labels plus notes; it is allowed to remain a template while the human audit is incomplete, but the readiness gate rejects unfilled fields once labels are complete. The finalization shortcut can import the wide worksheet with `--from-worksheet` or the model-blinded worksheet with `--from-model-blinded-worksheet`, run validation and finalization, rebuild the artifact package, and rerun the paper-aware readiness check after an independent human auditor has filled the labels and notes. The pre-release verifier runs the minimal example, test suite, paper build, readiness check, and git diff check before final release. In the current artifact package, the readiness report is intentionally **blocked** until the 50-row active-verification sheet is independently human-labeled and finalized. The Codex-assisted audit files are retained as triage artifacts and are not counted as human validation.